Practical No 5:

1. Write code to display a message when a button is clicked.

XML CODE:

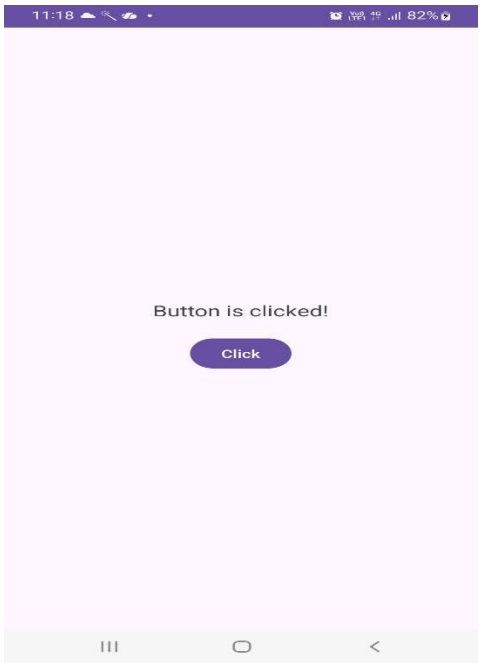
```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    tools:context=".MainActivity">

    <TextView
        android:id="@+id/txtMsg"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_margin="20dp"
        android:textSize="20dp"
        android:text="Welcome"
         />

    <Button
        android:id="@+id/btnClick"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Click"
        />

</LinearLayout>
```

JAVA CODE:

```
package com.example.appbuttontype;

import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;
```

```
import android.view.View;
import android.widget.TextView;
import android.widget.Button;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        TextView txt = findViewById(R.id.txtMsg );
        Button btnClick = findViewById(R.id.btnClick );

        btnClick.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                txt.setText("Button is clicked!");
            }
        });
    }
}
```

OUTPUT:



2. Create a layout with all three buttons: Button, ImageButton and ToggleButton.

XML CODE:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    tools:context=".MainActivity">

    <Button
        android:id="@+id/btnClick"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_margin="20dp"
        android:text="Click"   />

    <ImageButton
        android:id="@+id/imageButton"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_margin="20dp"
        app:srcCompat="@drawable/img" />

    <ToggleButton
        android:id="@+id/togBtn"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_margin="20dp"
        android:text="ToggleButton"
        android:textOn="WiFi ON"
        android:textOff="WiFi OFF" />

</LinearLayout>
```

JAVA CODE:

```
package com.example.appbuttontype;

import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;

import android.view.View;
import android.widget.TextView;
import android.widget.Button;

public class MainActivity extends AppCompatActivity {

   @Override
   protected void onCreate(Bundle savedInstanceState) {
      super.onCreate(savedInstanceState);
      setContentView(R.layout.activity_main);
      }
}
```

OUTPUT:

:

3. Change TextView Based on Toggle.

XML CODE:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    tools:context=".MainActivity">

    <TextView
        android:id="@+id/txtMsg"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_margin="20dp"
        android:textSize="20dp"
        android:text="Welcome"   />

    <ToggleButton
        android:id="@+id/togBtn"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_margin="20dp"
        android:text="ToggleButton"
        android:textOn="WiFi ON"
        android:textOff="WiFi OFF" />
</LinearLayout>
```

JAVA CODE:

```
package com.example.appbuttontype;

import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;

import android.view.View;
```

```
import android.widget.CompoundButton;
import android.widget.TextView;

 import android.widget.ToggleButton;


public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        TextView txt = findViewById(R.id.txtMsg );
        ToggleButton btnToggle = findViewById(R.id.togBtn );

        btnToggle.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                if ( btnToggle.isChecked() ) {
                    txt.setText("WiFi is ON!");
                }
                else {
                    txt.setText("WiFi is OFF!");
                }
            }
        });
    }
}
```

OUTPUT:

4. ImageButton Interaction i)Add an ImageButton using an image from the drawable folder.ii)On click, display a Toast message: “Image Button Pressed”.

XML CODE:

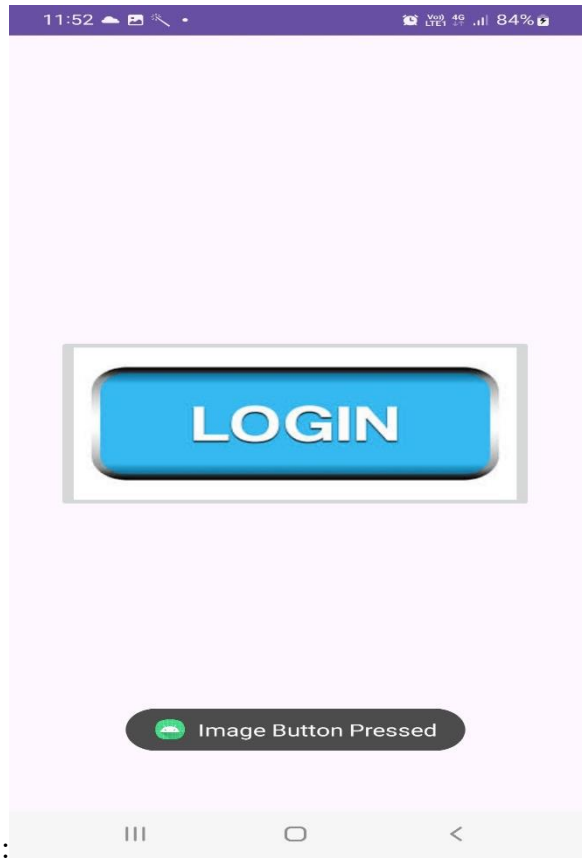
```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    tools:context=".MainActivity">

    <ImageButton
        android:id="@+id/imageButton"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_margin="20dp"
        app:srcCompat="@drawable/img" />

</LinearLayout>
```

JAVA CODE:

```
package com.example.appbuttontype;

import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;

import android.view.View;
import android.widget.ImageButton;
import android.widget.Toast;


public class MainActivity extends AppCompatActivity {
```

```
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);


        ImageButton btnImg = findViewById(R.id.imageButton );

        btnImg.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                Toast.makeText(getApplicationContext(), "Image Button Pressed",
Toast.LENGTH_LONG).show();
            }
        });
    }
}
```

OUTPUT



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